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Enhancing Public Safety through Real-time Weapon Detection: A Deep Learning Approach

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***Abstract*—Date In response to the imperative need for mitigating criminal activities and ensuring public safety, this research proposes a novel approach leveraging deep learning techniques for real-time weapon detection. In contemporary society, criminal acts pose significant threats to both individuals and the broader community, necessitating proactive measures to counter such offenses swiftly. By harnessing advanced technologies, particularly deep learning algorithms like YOLOv7-tiny, this study intend to create a robust system capable of identifying weapons in surveillance footage captured by CCTV cameras. The project begins with the meticulous collection of datasets from reputable research sources, ensuring diverse and comprehensive samples for training the deep learning model. Subsequently, these datasets undergo rigorous pre-processing steps to optimize them for training, including resizing images to the required format. To facilitate accurate annotation and labelling of the dataset, advanced tools like Robo flow software are employed, streamlining the process and enhancing efficiency. Upon completion of the pre-processing phase, the YOLOv7-tiny algorithm is deployed for training the model on the annotated datasets. Through iterative training and validation processes, the model is fine-tuned to achieve optimal performance in weapon detection tasks. Once trained, the model is capable of swiftly and accurately identifying weapons in real-time surveillance footage. This project presents an effective solution for real-time weapon detection in public spaces, leveraging deep learning algorithms and advanced software tools.**

Keywords— Dataset Collection, Pre-processing, Image Resizing, Annotated Images, Roboflow Software, Surveillance Footage, CCTV Cameras, Weapon Detection, Deep Learning etc.,

I. INTRODUCTION

Weapon or anomaly detection plays a crucial role in identifying irregular, unexpected, and unusual events or items that deviate from standard patterns. The impact of gun-related violence on public health, psychology, and the economy is profound, with numerous deaths and psychological traumas reported annually, especially among children exposed to violence. Addressing such issues requires proactive measures, including early detection of disruptive behavior and careful monitoring of suspicious learning algorithms and advanced image processing techniques.

Recent advancements in technology, including the availability of large datasets, faster GPUs, and sophisticated algorithms, have enabled the development of highly accurate automated detection systems. Machine learning algorithms coupled with smart surveillance devices have revolutionized the field, offering improved detection and tracking capabilities. However, real-time, online tracking of human objects and weapons remains a computational challenge, often requiring centralized processing at cloud centers.

Efforts to address this challenge have focused on enhancing the reliability and speed of robot manipulators for real-time tracking applications. Future research directions include optimizing algorithms for online tracking, leveraging edge computing for decentralized processing, and integrating sensor fusion techniques to enhance detection accuracy.

In conclusion, the convergence of machine learning, image processing, and smart surveillance technologies holds immense promise for advancing weapon and anomaly detection systems. By leveraging these technologies effectively, we can enhance public safety, mitigate the impact of gun-related violence, and create safer communities globally.

This study's organizational framework separates the research into several areas. Section 2 presents the Literature Survey. Section 3 covered the suggested system techniques. Additionally, results are addressed and illustrated in section 4. Section 5 concludes with a presentation of future work and conclusions.

II. LITERATURE SURVEY

"Deep Learning-Based Object Detection for Surveillance Systems" by Zhang et al. (2018). It discusses various architectures and algorithms used for real-time detection and highlights challenges and future research directions.

"Anomaly Detection in Surveillance Video: A Review" by Ahmed et al. (2020). This paper discusses different datasets, evaluation metrics, and challenges in anomaly detection, offering insights into potential solutions.

"Gun Detection in Surveillance Videos Using Convolutional Neural Networks" by Wang et al. (2019). This work propose a CNN-based approach for accurate and efficient detection of firearms, discussing dataset preparation, network architecture, and evaluation metrics.

"Real-Time Weapon Detection System using Deep Learning" by Singh et al. (2021). It discusses the object detection using algorithm, specifically tailored for detecting weapons in surveillance footage. Evaluation results and practical implementation aspects are also discussed.

"A Survey of Deep Learning Techniques for Object Detection and Classification in Surveillance Systems" by Gupta et al. (2020) Gupta et al. provide a comprehensive survey of deep learning techniques for object detection and classification in surveillance systems. The paper covers various architectures, datasets, evaluation metrics, and challenges, offering a holistic view of the state-of-the-art in this field.

"Review on Surveillance System for Weapon Detection Using Deep Learning Techniques" by Sharma et al. (2020) . The paper discusses different architectures, datasets, and evaluation metrics, highlighting the strengths and limitations of existing approaches and suggesting areas for further research.

"An Overview of Anomaly Detection Techniques: Existing Solutions and Latest Trends" by Li et al. (2019). The paper discusses various applications, challenges, and future directions in anomaly detection, offering valuable insights for researchers and practitioners.

"A Comprehensive Survey of Deep Learning Techniques for Object Detection in Video Surveillance" by Chen et al. (2021). The paper covers various architectures, datasets, evaluation metrics, and challenges, providing a thorough understanding of the current landscape and future research directions.

III. Existing System

Security and safety are crucial for a country's economic strength. CCTV cameras are used for surveillance but still need human monitoring. Automated systems are needed for detecting illegal activities. Despite advancements, real-time weapon detection is challenging. Detecting weapons in CCTV footage is challenging due to angle differences and occlusions. This work aims to enhance security by using deep learning algorithms on CCTV footage. The researchers created a dataset using various sources since no standard dataset was available. They implemented binary classification with a focus on reducing false positives and false negatives. Two approaches, sliding window/classification and region proposal/object detection, are used. Various algorithms like VGG16, Inception-V3, Inception-ResNetV2, SSDMobileNetV1, FRIRv2, YOLOv3, and YOLOv4 are employed for weapon detection in CCTV footage.

IV. Proposed Method

The project uses YOLOv7-tiny, an advanced Deep Learning algorithm, for weapon detection in CCTV surveillance videos. The dataset is collected from various sources and split into training and testing sets. Data pre-processing is applied before training the model. YOLOv7-tiny is known for its innovative approach and capabilities, gaining attention in computer vision and machine learning communities. The project uses a deep learning algorithm to validate and evaluate datasets, focusing on detecting weapons like guns. A React JS mobile app provides real-time notifications for weapon detection, enabling quick response. The system efficiently predicts and detects weapons in public places, enhancing security measures.

A. System Architecture

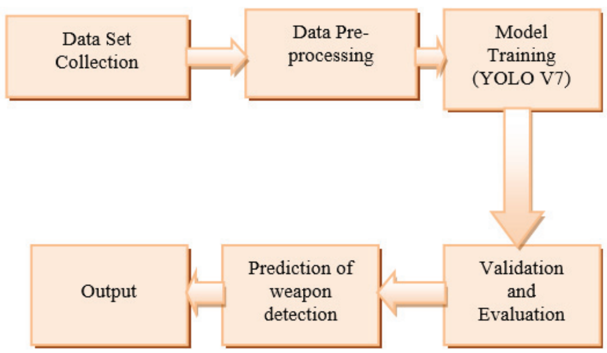

Fig. 1. System Architecture

1. **Dataset Collection**: Collecting datasets from various resources containing images or videos of CCTV camera surveillance footage depicting weapons.
2. **Data Splitting**: Separating the collected dataset into training and testing datasets. The testing dataset is kept separate and untouched for evaluation, ensuring the model's performance is accurately assessed.
3. **Data Pre-processing**: Image datasets are pre-processed by resizing, normalizing pixel values, and augmenting data to enhance quality and diversity, making them suitable for training the model.
4. **Model Training**: Utilizing the YOLOv7-tiny algorithm, a deep learning model is trained on the pre-processed dataset. The YOLOv7 algorithm is renowned for its efficiency and accuracy in object detection tasks, making it suitable for weapon detection in surveillance videos.
5. **Validation and Evaluation**: Validating and evaluating the trained model using the testing dataset to assess its performance metrics such as precision, recall, and accuracy. This step ensures that the model generalizes well to unseen data and effectively detects weapons in surveillance footage.
6. **Mobile Application Development**: Developing a mobile application using React JS to facilitate real-time notification on weapon detection. When a weapon, such as a gun, is detected in the surveillance footage.
7. **Real-time Prediction**: Integrating the trained model with the mobile application to enable real-time prediction of weapon detection via CCTV cameras. Users can input images or videos for prediction, and

the system swiftly identifies and alerts on the presence of weapons in public places, enhancing safety and security.

B. *Implementation*

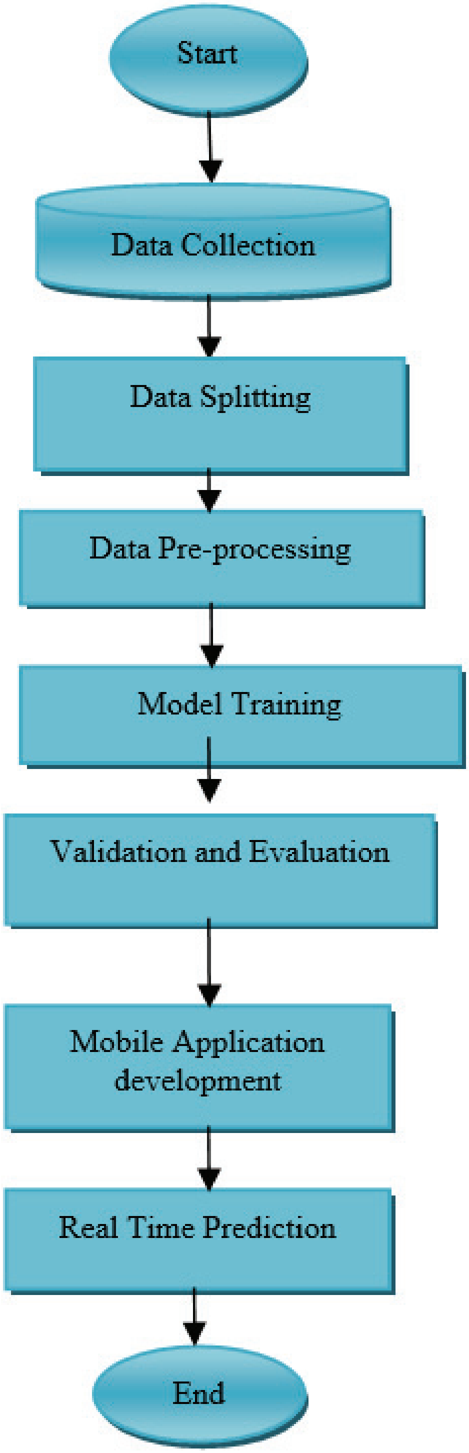

Fig. 2. Implmentation Flow Diagram

Flow chart illustrates the successive processes that make up the implementation process. Data loading and pre-processing: In this first stage, the dataset is loaded and the photos are pre-processed to make them ready of

1. **Dataset Collection**: Data is gathered from various sources containing CCTV camera surveillance footage depicting weapons.
2. **Data Splitting**: The collected dataset can be classified as training / testing datasets for model development and evaluation, respectively.
3. **Data Pre-processing**: The training dataset undergoes pre-processing to enhance quality and suitability for model training.
4. **Model Training**: The pre-processed dataset is used to train the YOLOv7-tiny algorithm, for weapon detection using deep learning model.
5. **Validation & Evaluation**: The model which are trained is evaluated using the testing dataset to assess its performance metrics.
6. **Mobile Application Development**: A mobile application is developed using React JS to facilitate real-time notification on weapon detection.
7. **Real-time Prediction**: The trained model is integrated with the mobile application to enable.

V. Results and Discussion

We can divide our project into completed implementation modules to start with the testing of the trained model. Gathering image caption datasets is a step in the dataset collection process. The dataset was gathered for the project, and the following figure is displayed.

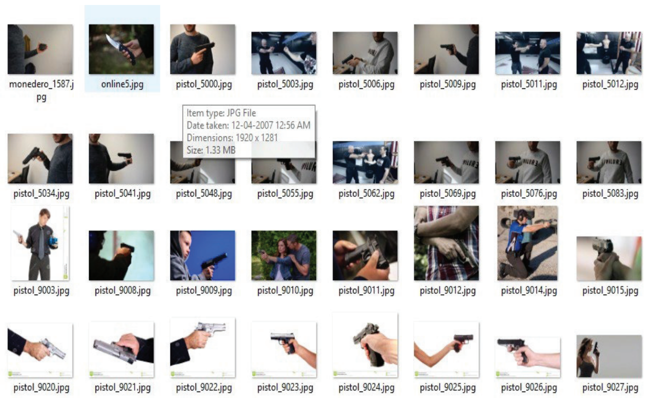

Fig. 3. Dataset Collection

After that these datasets undergo pre-processing to transform the photos into the necessary file sizes so that the model can be trained on them. The straightforward pre-processing methods for image resizing are depicted in the figure below.

Fig. 4. Pre-processing

The image annotated using roboflow software is seen in the figure below.

Fig. 5. Annotated image

The experiment results for Figure 3 Dataset Collection indicate the successful acquisition of datasets from various sources. This stage involved gathering images or videos from CCTV camera surveillance footage depicting weapons learning algorithm used for weapon detection.

Figure 4 depicts the pre-processing stage, specifically showcasing the simple techniques employed for image resizing. These techniques ensure that the images are standardized to a uniform size, facilitating efficient training and optimizing computational resources.

Once the datasets are pre-processed, they are ready for model training. However, before proceeding with training, it is essential to annotate the images to provide ground truth labels for the objects of interest—in this case, weapons. Figure 5 displays an annotated image using Roboflow software, where weapons are delineated and labeled for training the model.

The dataset collection process ensures that a diverse and representative set of data is available for subsequent model training and evaluation.

Following dataset collection, the datasets undergo pre-processing to prepare them for training with the model. This involves transforming the photos into the necessary file sizes and formats, ensuring compatibility with the deep

VI. Conclusion And Future Scope

Using an advanced deep learning algorithm, the project has successfully been implemented to give a solution for weapon detection. Yolov7-tiny is one algorithm that predicts weapon detection more accurately. The best method for ensuring accuracy in the output is deep learning, which produces guessing of the greatest caliber. There is a great deal of room for technological advancement because there are numerous ways to identify weaponry.

VII. Future Scope

In the near future, we will examine how weapon detection is applied in diverse industries to identify technological advancements in the detection domain that can support more accurate auto alert systems and secure cities. Therefore, this concept has a promising future when manual prediction may be inexpensively converted to computerized production.

References

[1] Briqech, Z., Gupta, S., Beltayib, A., Elboushi, A., Sebak, A. R., and Denidni, T. A. (2020). 57-64 GHz Imaging/Location Sensor — Part II: Investigates Covered Weapons and Undermining Materials Recognition. IEEE Sensors Diary, 1-1. doi:10.1109/jsen.2020.2997293

[2] Final plan Calculation for Directed Weapon Framework Against Elevated Dodger. IEEE Exchanges on Aviation and Electronic Frameworks (Volume: 58, Issue: 1, February 2022) DOI: 10.1109/TAES.2021.3117090

[3] Fath U Min Ullah, Khan Muhammad, Ijaz Ul Haq, Noman Khan, Ali Asghar Heidari, Sung Wook Baik , (2022). Artificial intelligence Helped Edge Vision for Savagery Recognition in IoT-Based Modern Observation Organizations. IEEE Exchanges ON Modern INFORMATICS, VOL. 18, NO. 8, AUGUST 2022. Advanced Item Identifier 10.1109/TII.2021.3116377

[4] Garcia-Rial, F., Montesano, D., Gomez, I., Callejero, C., Bazus, F., and Grajal, J. (2018). Joining Monetarily Accessible Dynamic and Latent Sensors Into a Millimeter-Wave Imager for Disguised Weapon Recognition. IEEE Exchanges on Microwave Hypothesis and Strategies, 1-17. doi:10.1109/tmtt.2018.2880757

[5] T.Sripriya, Dr.V.Jeyalakshmi, "Simulation of Different Diaphragm Structures for Optical MEMS Pressure Sensors and Its Comparison", International Journal of Applied Engineering Research, ISSN 0973-4562 Volume 10, Number 5 PP. 11749-11754, 2015.

[6] Madhumita Chakravarti; Santanu Bijoy Majumder; G. Shivana, Sun, Y., and Tooth, Z. (2021). Research on Projection Dark Objective Model In light of FANP-QFD for Weapon Arrangement of Frameworks Capacity Assessment. IEEE Frameworks Diary, 15(3), 4126-4136. doi:10.1109/jsyst.2020.3027585

[7] Thomas Reinhold; Christian Reuter. (2021). Toward a Digital Weapons Evaluation Model — Evaluation of the Specialized Highlights of Pernicious Programming. IEEE Exchanges on Innovation and Society (Volume: 3, Issue: 3, September 2022), DOI: 10.1109/TTS.2021.3131817

[8] Tivive, F. H. C., Bouzerdoum, A., and Abeynayake, C. (2018). GPR Target Location by Joint Scanty and Low-Rank Framework Disintegration. IEEE Exchanges on Geoscience and Remote Detecting, 1-13. doi:10.1109/tgrs.2018.2875102

[9] Vavilov, S. A., and Lytaev, M. S. (2019). Displaying condition for numerous blade edge diffraction. IEEE Exchanges on Recieving wires and Proliferation, 1-1. doi:10.1109/tap.2019.2957085

[10] T.Sripriya, Dr.V.Jeyalakshmi, "Study of Rectangular diaphragm - Microelectromechanical Optical pressure sensor", Journal of Computational and Theoretical Nanoscience, ISSN 1546-1955, DOI: 10.1166/jctn.2017.6641, Vol.14.No.7 PP.3399-3404 July 2017.